

Tianyi Zhou

✉ tzhou25@seas.upenn.edu | 📞 +1.312.622.3218 | 🌐 goldenexpn.github.io | 🐙 GoldenEXPn

RESEARCH INTERESTS

Machine learning and artificial intelligence, with a focus on external memory systems for LLM, continual and parameter-efficient learning for foundation models, reinforcement learning, and representation learning.

EDUCATION

2025–Present	University of Pennsylvania , M.S.E. in Systems Engineering	GPA: 3.57/4.0
2021–2025	University of Rochester , Computer Science and Digital Media Studies	GPA: 3.70/4.0

RESEARCH EXPERIENCE

Research Assistant – Ale Lab, University of Pennsylvania Oct. 2025–Present

- Designed and implemented **Code Forest**, a structured external-memory framework for LLM code generation that organizes programming problems, composite functions, atomic modules, and implementations into a multi-resolution knowledge forest.
- Built an end-to-end construction and retrieval pipeline combining **LLM distillation, AST-based caller–callee analysis, semantic embedding search, LLM-guided module/tag merging, and layer-wise relevance tree-like traversal**; implemented both offline forest construction and online incremental updates.
- Developed reproducible execution-based evaluation infrastructure with frozen gold-passing test suites, paired Vanilla/RAG scoring, retrieval ablations, and audit tools; a strict paired pilot improved pass-all accuracy from **55.2% to 62.1% (+6.9 pp)** and mean partial score from **71.7% to 80.0%**.
- Built a canonical execution-evaluation population of **189 held-out problems** with at least five validated test suites per eligible problem and **1,998 retained gold-passing test inputs**, enabling controlled large-scale retrieval experiments and future learned navigation policies.

Research Assistant – Huang Lab, University of Pennsylvania Feb. 2026–Present

- Designed a supervised and continual-learning benchmark for text-driven pathology foundation models across **41 datasets and 259 pathology classes**.
- Benchmarked LiteFM-, BioMedCLIP-, and CONCH-based systems under full/partial fine-tuning, LoRA, replay, knowledge distillation, EWC, and recent adapter-based continual-learning methods, reaching **0.911 mean balanced accuracy** on the 37-dataset base benchmark.
- Established **replay + knowledge distillation** as the strongest shared-adapter baseline in the current benchmark, retaining **0.910 balanced accuracy** on previously learned datasets while adapting to four new pathology tasks.
- Performed LoRA weight-space and subspace analyses to characterize adapter interference and catastrophic forgetting, motivating **PathCAR**, a prompt-conditioned continual adapter-routing framework under development.

INDUSTRY EXPERIENCE

ECCOM Network System Co. – Machine Learning Engineer Intern May 2024–Aug. 2024

- Developed and evaluated feature-selection and importance-analysis workflows aligned with client requirements for tabular business data.
- Built classification pipelines with **XGBoost, CatBoost, and TabNet**, including preprocessing, feature engineering, model comparison, and interpretation.
- Designed an end-to-end employee-turnover prediction workflow achieving **90%+ accuracy** on a blind test set.

Red Note (Xiaohongshu) – Business Intelligence Intern May 2023–Aug. 2023

- Built web-scraping pipelines to track competitor e-commerce sales performance and pricing behavior.
- Applied **TF-IDF** and keyword analysis to user/content text for market and brand intelligence.
- Developed an **LSTM-based purchase-intent model** from historical user activity signals.

SELECTED PROJECTS

LLM Subtitle Translation Pipeline

2026

- Built a resumable five-stage media pipeline (**extract** → **transcribe** → **translate** → **verify** → **mux**) integrating FFmpeg, Whisper, and pluggable OpenAI-compatible LLM APIs.
- Designed glossary-constrained translation, index-keyed structured outputs, missing-cue fallback, byte-preserving timestamp handling, and automated verification to prevent common LLM ordering and omission failures.
- Re-translated **13 episodes / 5,248 subtitle cues**; glossary and integrity controls reduced the empty-line rate on a representative episode from **16.1% to 0%**.

Image2GPS – Vision-Based Geolocation

2026

- Developed a campus-scale image geolocation model using a pretrained **MobileNetV3-Small** backbone with learnable **GeM pooling** and a **24×20 spatial grid**.
- Formulated localization as joint grid-cell classification and cell-specific coordinate regression, decoding predictions as a probability-weighted mixture of candidate coordinates.
- Combined cross-entropy, offset regression, and **Haversine-distance** objectives and built an evaluator for geodesic error, MAE/RMSE, and inference latency.

PPO Locomotion and Cross-Task Policy Transfer

2026

- Implemented and iteratively tuned a **PPO** training workflow for DeepMind Control walker stand/walk tasks with observation normalization, KL-aware policy updates, reward diagnostics, checkpointing, and separate raw/shaped-return tracking.
- Improved walker-stand raw episodic return from approximately **128 to 858** over 4.1M environment steps; transferred the learned stand policy and observation normalizer to the walk task while reinitializing the critic for cross-task adaptation.

Benchmarking Parameter-Efficient Fine-Tuning for LLMs

2024

- Compared LoRA-family methods including **LoRA-FA**, **AdaLoRA**, and **VeRA** on Llama-3-8B and studied quality/efficiency trade-offs.
- Built an automated LLM-based evaluation workflow for model quality, latency, and memory-efficiency analysis on HPC resources.

TEACHING EXPERIENCE

Recitation Teaching Assistant – Computer Models and Limitations

Jan. 2025–May 2025

Workshop Teaching Assistant – Intro to Programming

Aug. 2024–Dec. 2024

Teaching Assistant – Computational Statistics

Aug. 2023–May 2024

- Led weekly workshop sessions and provided one-on-one assistance in **R and Python**; graded assignments and exams.

TECHNICAL SKILLS

Languages	Python, C/C++, R, SQL, Bash
ML / AI	PyTorch, Hugging Face Transformers/PEFT, Scikit-learn, XGBoost, Pandas, NumPy
Research	LLM reasoning and retrieval, RAG, continual learning, LoRA/PEFT, knowledge distillation, reinforcement learning, AST analysis, computer vision
Infrastructure	Linux, Git, SLURM/HPC, Docker, Weights & Biases, FFmpeg